



SMT. SUDHATAI MANDKE COLLEGE

**AFFILIATED TO SAVITRIBAI PHULE PUNE UNIVERSITY
APPROVED BY GOVERNMENT OF MAHARASHTRA
ID NO. PU / PN / ASC / 178 - 2002**

A PROJECT REPORT ON

AI RESUME ANALYZER

SUBMITTED TO

**SAVITRIBAI PHULE PUNE UNIVERSITY
BY**

MR. ADITYA MAHADEO PAWAR

**BACHELOR OF COMPUTER APPLICATION
SEM-VI (2025-26)**

UNDER GUIDANCE OF

PROF. AMBARA YANDE

47/8, MANDKE GROWTH CENTRE

PAUD ROAD, PAUD PHATA FLYOVER, PUNE-411038

WEBSITE: WWW.MANDKECOLLEGE.NET

EMAIL: MANDKECOLLEGE@GMAIL.COM

INDEX

Sr No.	Title
1.	Abstract
2.	Acknowledgments
3.	Introduction ● Motivation ● Problem Statement ● Purpose/Objective and Goals ● Literature Survey ● Project Scope and Limitations
4.	System Analysis ● Existing System ● Scope and Limitations of Existing Systems ● Project Perspective, Features ● Stakeholders ● Requirements Analysis – Functional requirements, Performance requirements, Security requirements etc..
5.	System Design ● Design Constraints ● System Model: UML Diagram ● User Interfaces ● Data Dictionary
6.	Implementation Details ● Software/Hardware Specification
7.	Outputs and Reports Testing ● Test Plan, Black Box Testing or Data Validation Test Cases, White Box Testing or Functional Validation Test Cases and Results
8.	Conclusion and Reports Testing
9.	Future Scope
10.	Bibliography and Reference

❖ ABSTRACT

The **AI Resume Analyzer** is an intelligent web-based application designed to automate and enhance the process of resume screening and evaluation. In today's competitive job market, recruiters receive a large number of resumes for a single job position, making manual screening time-consuming and inefficient. This project aims to simplify and optimize the recruitment process using Artificial Intelligence and data analysis techniques.

The system allows users to upload resumes in digital format (PDF/DOC). It then analyses the content using Natural Language Processing (NLP) to extract key information such as skills, education, experience, and keywords.

The application provides a score-based analysis, highlights missing skills, and suggests improvements to enhance the resume. It can also categorize resumes based on different job roles such as software developer, data analyst, and business analyst. The backend is implemented using Python, while the database is managed using MySQL (via XAMPP), ensuring efficient data storage and retrieval.

❖ ACKNOWLEDGEMENTS

I take this space to acknowledge and extend my heartfelt gratitude to those who have helped in various ways through the project work to make this project a success.

First, I wish to express my gratitude to SMT. SUDHATAI MANDAKE COLLEGE for giving me the opportunity to apply the knowledge gained through the degree program.

I also feel thankful and express my kind gratitude towards our Principal Sir for allowing me to conduct the College Management Project. The mentioned project was done under the guidance of Mrs. Ambara Yande. I feel thankful to the BBA(CA) staff for giving me such a big opportunity.

❖ INTRODUCTION

The AI Resume Analyzer System is a web-based application designed to analyze and evaluate resumes using Artificial Intelligence. In traditional recruitment systems, recruiters manually read and shortlist resumes, which is time-consuming, requires a lot of effort, and may lead to human errors or biased decisions. To overcome these problems, the AI Resume Analyzer is developed to automate the resume screening process.

This system provides two main modules: Admin module and User module. The Admin can manage job roles, set required skills, and view uploaded resumes along with analysis results. The User can upload their resume, view their resume score, and get suggestions for improvement. The system automatically extracts important information such as skills, education, and experience from the resume and compares it with job requirements.

The system is developed using Python (Flask), MySQL, HTML, CSS, JavaScript, and Bootstrap. The database stores user details, resumes, extracted data, and analysis results. The system also includes features like skill matching, keyword analysis, and resume scoring.

The main objective of this system is to make the resume screening process smart, automated, and efficient. It helps in reducing manual work, saving time, improving accuracy, and helping users enhance their resumes according to industry standards.

❖ MOTIVATION

In today's competitive job market, a single job opening receives a large number of resumes, making it difficult for recruiters to manually screen each application efficiently. The traditional method of resume evaluation is time-consuming, requires significant human effort, and may result in errors or biased decisions. Many deserving candidates may get rejected due to improper resume formatting or missing keywords.

At the same time, students and job seekers often struggle to create effective resumes that meet industry standards. They are unaware of the skills and keywords required for specific job roles, which reduces their chances of getting shortlisted.

This motivated the development of the **AI Resume Analyzer System**, which automates the resume screening process and provides accurate analysis using Artificial Intelligence. The system helps recruiters by reducing workload and improving efficiency, while also guiding users to improve their resumes with proper suggestions.

The goal of this project is to create a smart, fast, and reliable system that benefits both recruiters and job seekers by making the hiring process more transparent, efficient, and data-driven.

❖ PROBLEM STATEMENT

In the current recruitment process, organizations receive a large number of resumes for different job roles. Screening these resumes manually is a time-consuming and tedious task for recruiters. It requires a lot of effort to read each resume, identify relevant skills, and shortlist suitable candidates. This process may also lead to human errors, inconsistency, and biased decision-making.

Additionally, many job seekers are not aware of proper resume formats and industry-required skills. Due to missing keywords or improper presentation, their resumes may get rejected even if they are capable candidates.

There is no efficient system available to automatically analyse resumes, compare them with job requirements, and provide accurate feedback to both recruiters and candidates.

Therefore, there is a need to develop an AI-based Resume Analyzer System that can automate the resume screening process, reduce manual effort, improve accuracy, and provide useful suggestions for resume improvement.

❖ PURPOSE OF THE PROJECT

The purpose of the **AI Resume Analyzer System** is to develop an intelligent and automated solution for analysing and evaluating resumes efficiently. The system aims to simplify the recruitment process by reducing the manual effort involved in screening large numbers of resumes.

The project is designed to help recruiters quickly identify suitable candidates by automatically extracting important information such as skills, education, and experience from resumes. It compares this data with job requirements and generates a score-based evaluation, making the selection process faster and more accurate.

At the same time, the system also helps job seekers by providing feedback and suggestions to improve their resumes according to industry standards. This increases their chances of getting shortlisted for job opportunities.

Overall, the purpose of this project is to create a smart, time-saving, and reliable system that enhances the efficiency of the hiring process and benefits both recruiters and candidates.

❖ LITERATURE SURVEY

The recruitment process has evolved significantly with the advancement of technology, especially with the introduction of Artificial Intelligence and automation tools. Traditional resume screening methods involve manual evaluation, which is time-consuming, inefficient, and prone to human errors. To overcome these limitations, various research works and systems have been developed in the field of automated resume analysis.

Many existing systems use **Natural Language Processing (NLP)** techniques to extract important information such as skills, education, and experience from resumes. These systems analyse textual data and identify relevant keywords that match job descriptions. Some applications also use **Machine Learning algorithms** to improve the accuracy of candidate selection by learning from past data.

Research studies have shown that automated resume screening systems can significantly.

Some systems also focus on resume scoring and ranking based on job requirements, but they may not be fully user-friendly or accessible for students and freshers. Additionally, there is a need for a system that not only analyzes resumes but also provides suggestions to improve them according to current industry standards.

Based on the existing studies and systems, the proposed AI Resume Analyzer aims to combine resume analysis, scoring, and improvement suggestions in a single platform. The system focuses on accuracy, ease of use, and efficiency, making it beneficial for both recruiters and job seekers.

❖ PROJECT SCOPE

The scope of the AI Resume Analyzer System is to develop a web-based application that automates the process of resume analysis and evaluation using Artificial Intelligence techniques. The system is designed to be used by both recruiters and job seekers for efficient resume screening and improvement.

The project focuses on analysing resumes in digital formats such as PDF and extracting important details like skills, education, and work experience. It compares this information with predefined job requirements and provides a score-based evaluation along with suggestions for improvement.

The system includes features such as resume uploading, keyword extraction, skill matching, and result generation. It also allows the admin to manage job roles, required skills, and view candidate analysis reports. The database stores user data, resumes, and analysis results for future reference.

This project is mainly limited to text-based resume analysis and does not include advanced features like video interviews or real-time recruiter communication. The accuracy of the system depends on the quality of the resume content and predefined datasets.

In future, the system can be enhanced by adding advanced Machine Learning models, integration with job portals, and support for multiple languages to make it more powerful and scalable.

❖ LIMITATIONS

The AI Resume Analyzer System has certain limitations that may affect its performance and accuracy. The system mainly relies on keyword matching and predefined rules, so it may not fully understand the context or meaning of the content in a resume. As a result, some relevant candidates might not get accurate scores if their resumes do not contain expected keywords.

The system works only with digital resume formats such as PDF or DOC, and poorly formatted or scanned resumes may not be analysed correctly. The accuracy of the results also depends on the quality and structure of the uploaded resume.

Another limitation is that the system is designed for basic analysis and does not include advanced AI models or deep learning techniques, which could improve decision-making. It also does not support real-time interaction with recruiters or integration with external job portals.

Additionally, the system currently supports limited job roles and predefined skill sets. It may not perform well for highly specialized or uncommon job profiles.

Despite these limitations, the system provides a useful and efficient solution for basic resume screening and improvement.

❖ SYSTEM ANALYSIS

The system analysis of the AI Resume Analyzer focuses on understanding the limitations of the existing manual resume screening process and defining an improved automated solution. In the existing system, recruiters manually review resumes, which is time-consuming, inefficient, and prone to errors and bias. The scope of the existing system is limited to basic shortlisting without any automation, and it lacks accuracy and consistency when handling large numbers of resumes. The proposed system introduces a web-based platform that uses AI techniques to analyze resumes, extract skills, and match them with job requirements.

It includes features such as resume uploading, keyword extraction, scoring, and suggestion generation. The main stakeholders of the system are admin, recruiters, and job seekers who interact with the system for different purposes. The system requires functional capabilities like resume analysis and data storage, along with performance requirements for fast processing. Additionally, security requirements such as data protection and user authentication are also considered to ensure safe and reliable operation.

❖ EXISTING SYSTEM

In the existing recruitment system, resume screening is done manually by recruiters or HR personnel. They collect resumes from different sources such as emails and job portals and review each resume individually. This process is time-consuming and requires a lot of human effort, especially when there are a large number of applications.

The manual evaluation may also lead to errors, inconsistency, and biased decisions. There is no proper system for accurate skill matching or automatic filtering of candidates. Additionally, candidates do not receive any feedback or suggestions for improving their resumes. Hence, the existing system is inefficient and not suitable for modern large-scale recruitment needs.

❖ **SCOPE AND LIMITATIONS OF EXISTING SYSTEM**

- **SCOPE**

The scope of the existing system is limited to manual resume screening where recruiters review resumes and shortlist candidates based on their knowledge and experience. It is useful for basic recruitment processes and works well when the number of resumes is small. The system helps in identifying suitable candidates through human judgment but does not involve any automation or advanced technology.

- **LIMITATIONS**

The existing system has several limitations as it is time-consuming and requires a lot of manual effort. It is not suitable for handling a large number of resumes and may lead to human errors and biased decisions. There is no proper mechanism for skill matching or accurate evaluation of candidates. Additionally, the system does not provide any feedback or suggestions to candidates for improving their resumes, making the process less efficient.

❖ PROJECT PERSPECTIVE

The AI Resume Analyzer System is developed as a web-based application that aims to automate and improve the resume screening process. It acts as an independent system that can be used by both recruiters and job seekers for efficient resume analysis. The system provides a platform where users can upload resumes, and the application processes them using Artificial Intelligence techniques to extract and evaluate important information. It reduces the dependency on manual work and increases the speed and accuracy of candidate selection. The system can also be extended in the future by integrating it with job portals and advanced AI models to enhance its functionality and performance.

• FEATURES:

- **User Registration and Login:** Secure login system for users and admin.
- **Resume Upload:** Users can upload resumes in PDF/DOC format.
- **Resume Analysis:** System analyses resume content using AI techniques.
- **Skill Extraction:** Automatically extracts skills, education, and experience.
- **Keyword Matching:** Matches resume data with job requirements.
- **Resume Scoring:** Generates score based on resume quality and relevance.
- **Improvement Suggestions:** Provides tips to enhance the resume.
- **Admin Management:** Admin can manage job roles and required skills.
- **Data Storage:** Stores user data, resumes, and analysis results.
- **User-Friendly Interface:** Simple and easy-to-use interface for smooth operation.
- **Security:** Ensures data privacy and secure access to the system.

❖ STAKEHOLDERS

- **Admin:** The admin is responsible for managing the overall system. They handle job roles, define required skills, manage user data, and monitor resume analysis results. They ensure the smooth functioning and maintenance of the system.
- **Recruiters:** Recruiters are the main users who utilize the system to analyse resumes and shortlist candidates. The system helps them save time by automatically evaluating resumes and providing accurate results for better hiring decisions.
- **Job Seekers/Students:** These are the users who upload their resumes to the system. They use the platform to get their resumes analysed, receive scores, and obtain suggestions to improve their resumes according to industry standards.
- **Organizations/Companies:** Companies can use the system to streamline their recruitment process. It helps them efficiently handle a large number of applications and select suitable candidates based on required skills.
- **Technical Team:** Developers and IT professionals are responsible for developing, maintaining, and updating the system. They ensure the system runs smoothly, fix technical issues, and improve features over time.

❖ REQUIREMENT ANALYSIS

Functional Requirements

- User registration and login
- Users can upload resumes (PDF/DOC)
- System extracts skills, education, and experience
- Analyse resumes using keyword matching
- Generate resume score
- Provide suggestions for improvement
- Admin can add, edit, and manage job roles and skills
- Store and retrieve resume data from database

Performance Requirements

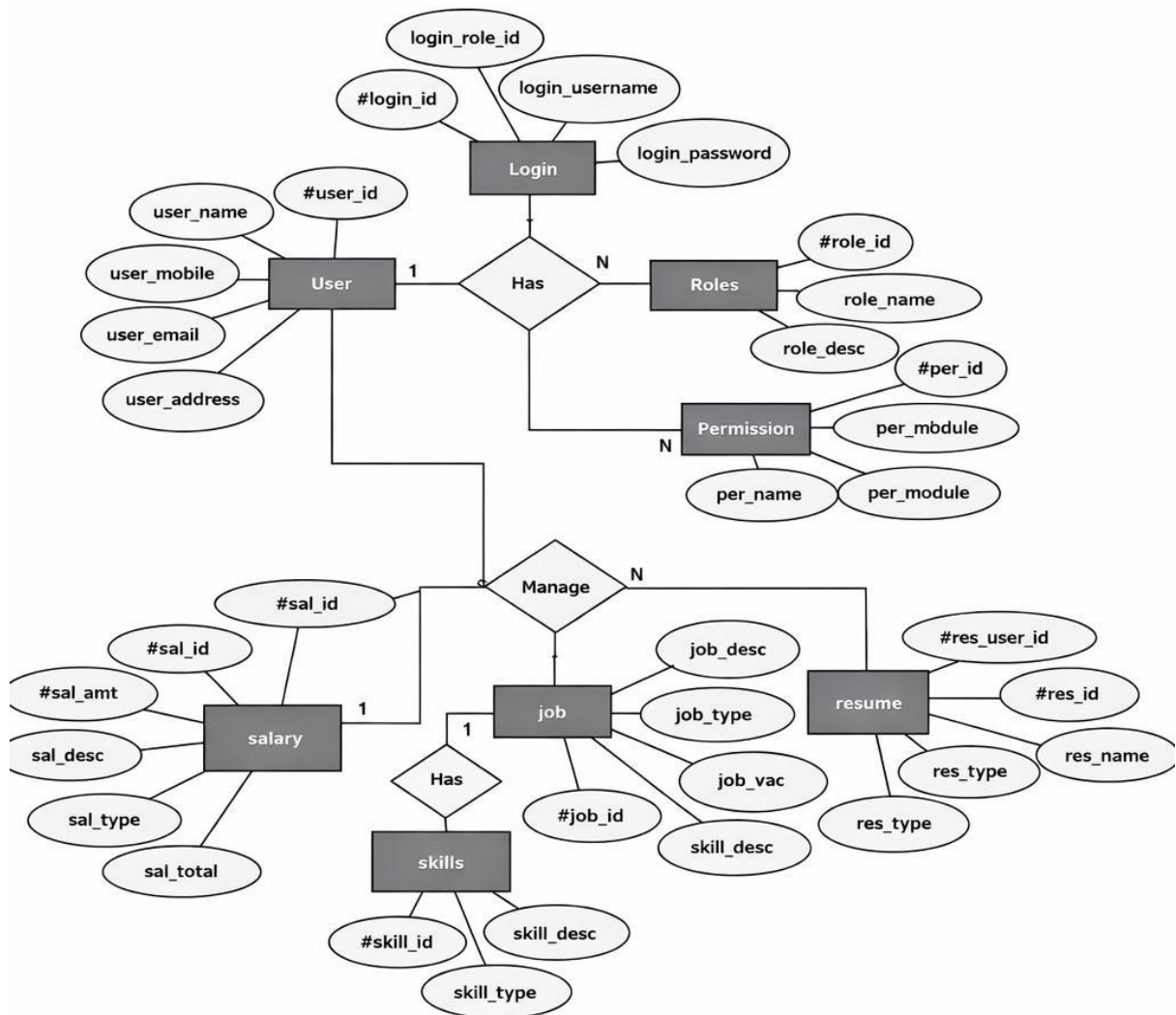
- Fast resume processing and analysis
- Ability to handle multiple users at the same time
- Efficient data extraction and storage
- Quick response time for results

Security Requirements

- Secure login and authentication system
- Protection of user data and resumes
- Prevention of unauthorized access
- Safe storage of sensitive information

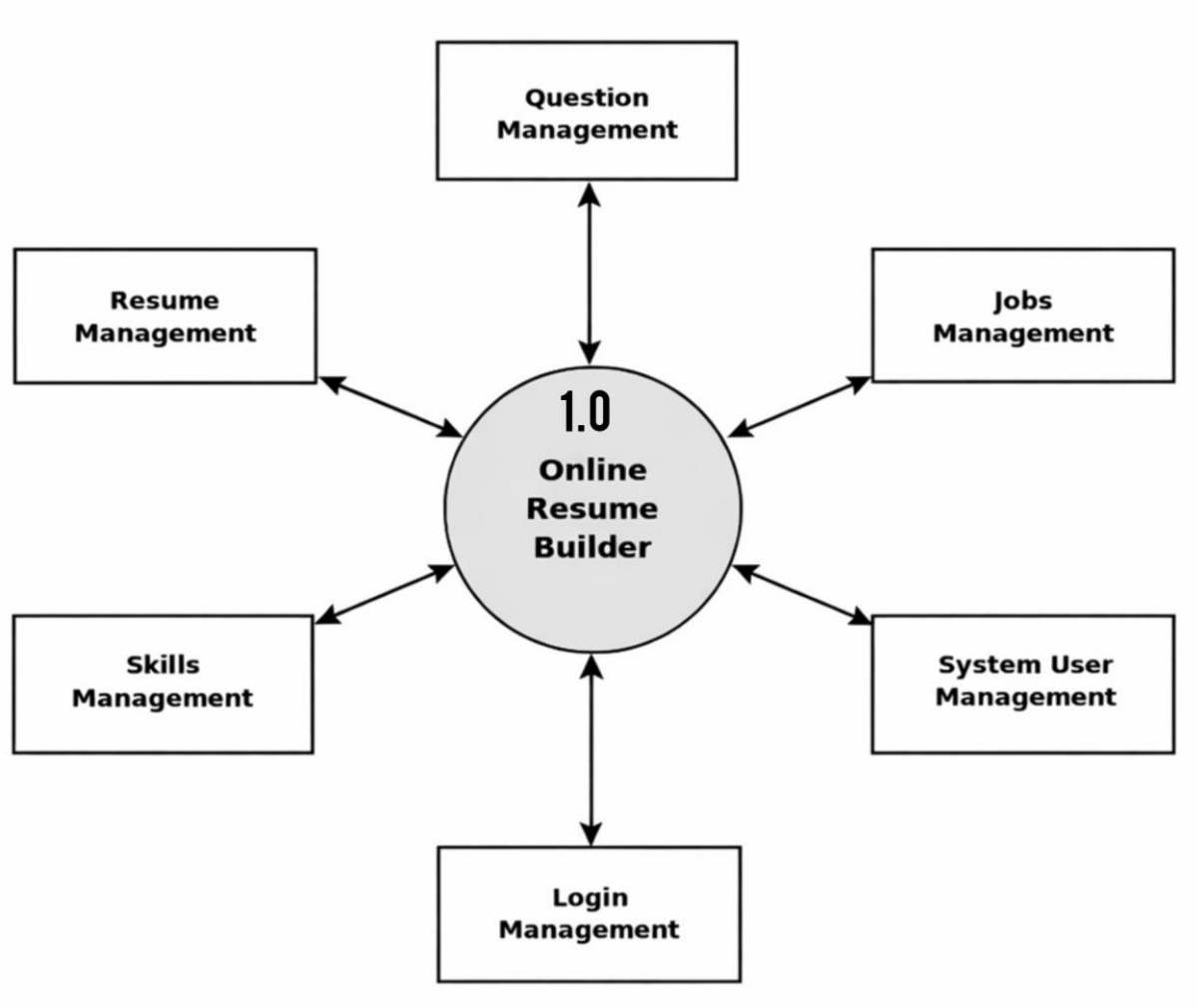
❖ SYSTEM DESIGN

ERD DIAGRAM

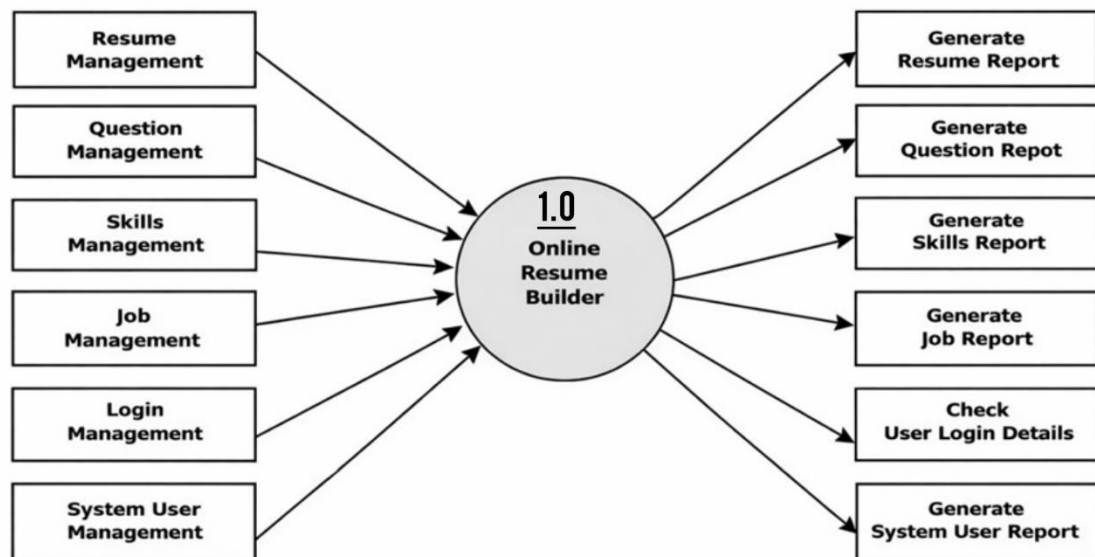


❖ DFD'S

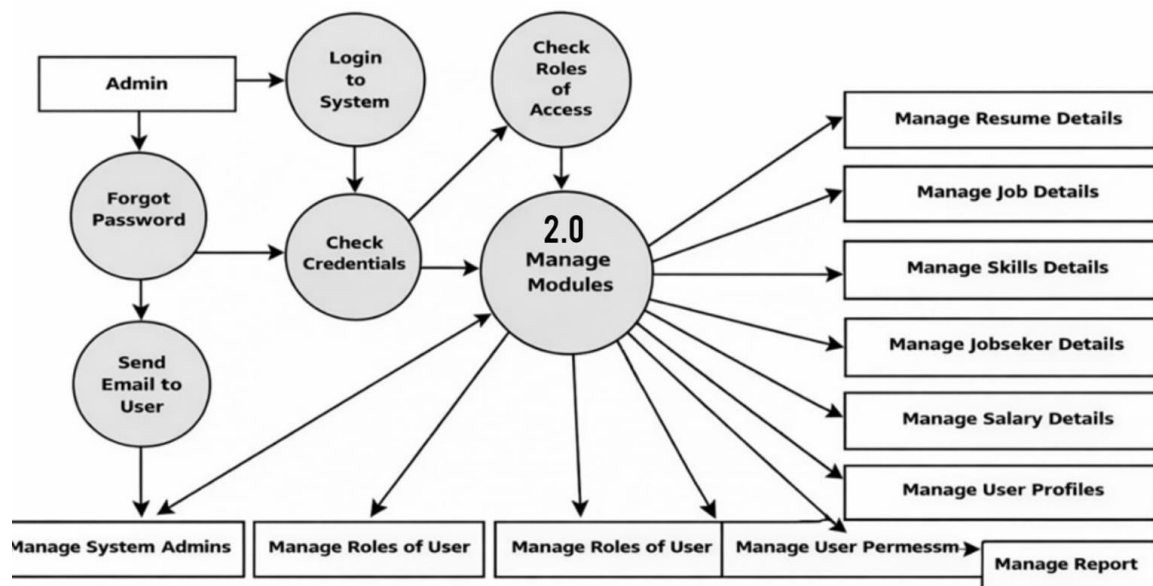
1.LEVEL 0 DFD



DFD LEVEL 1 –

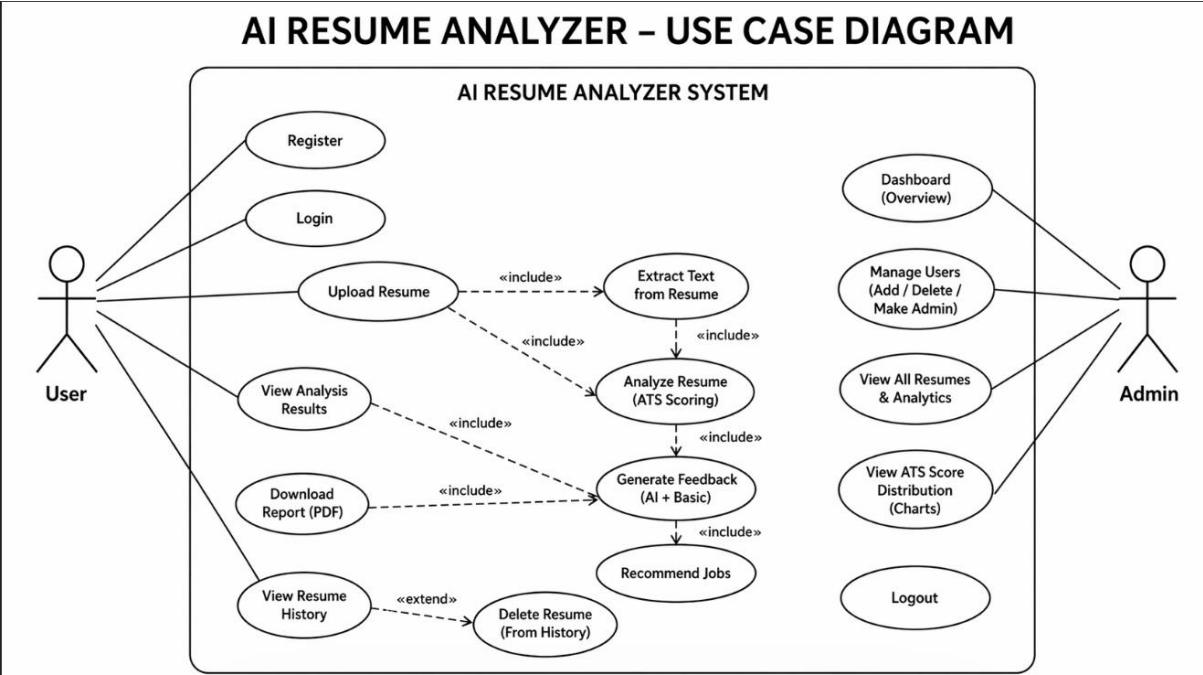


DFD LEVEL 2 –

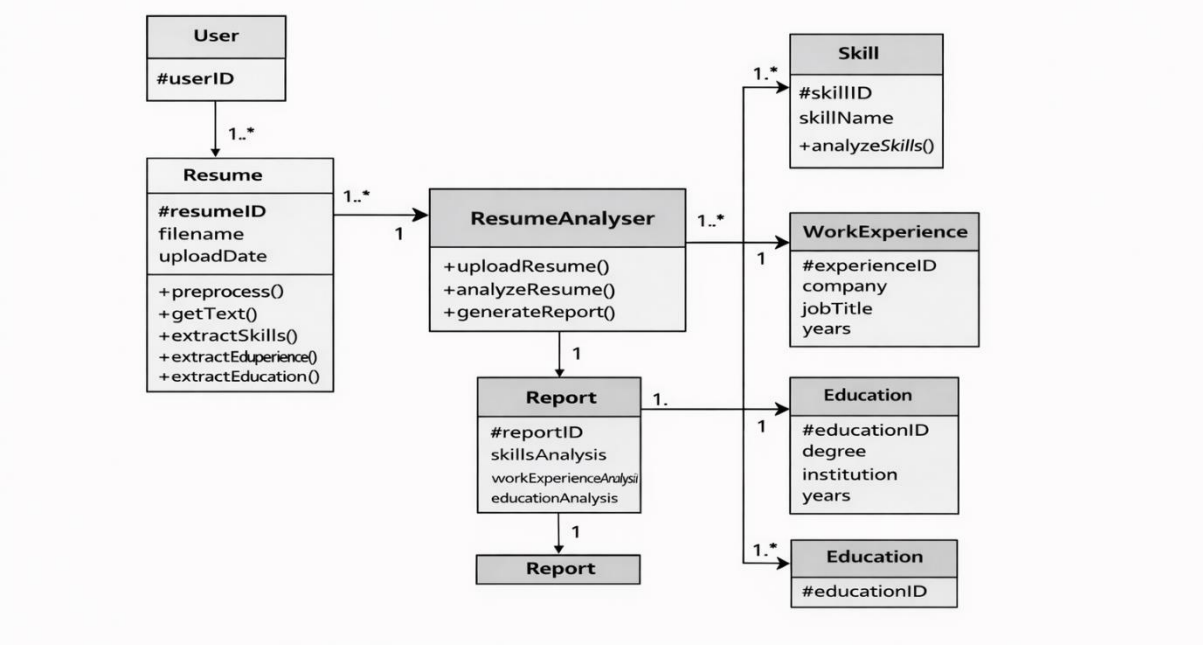


USE CASE DIAGRAM

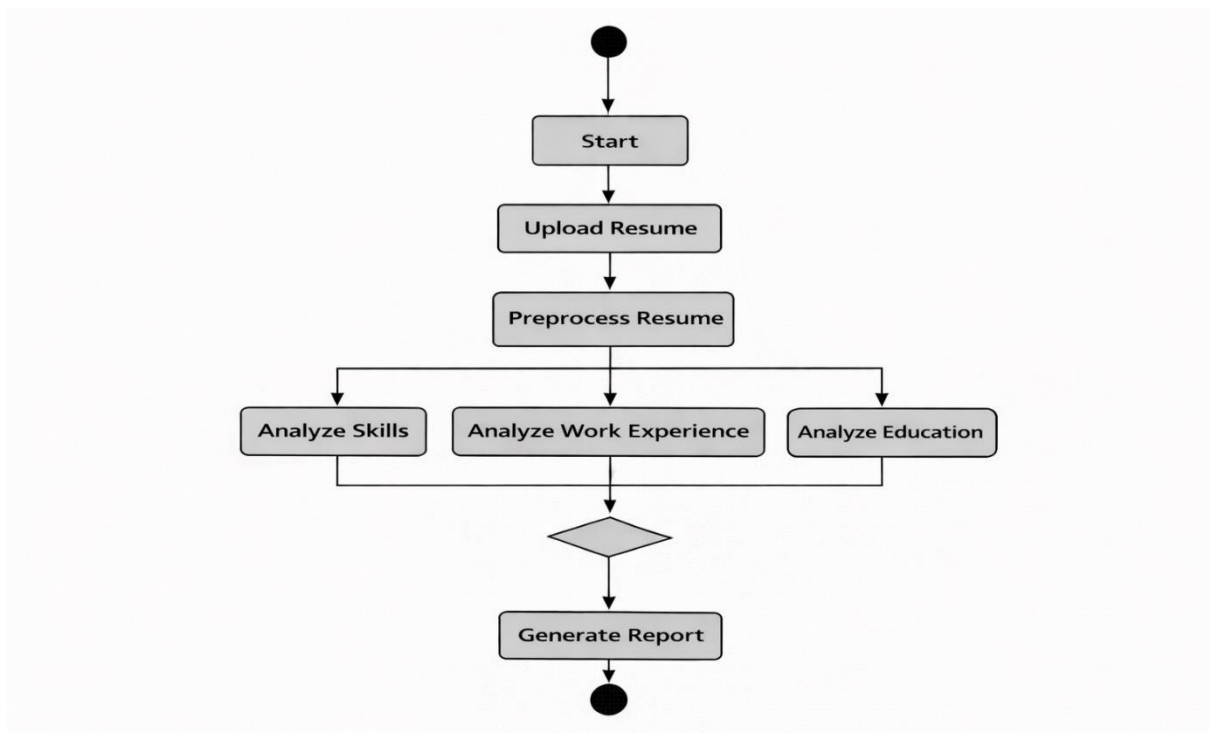
cre



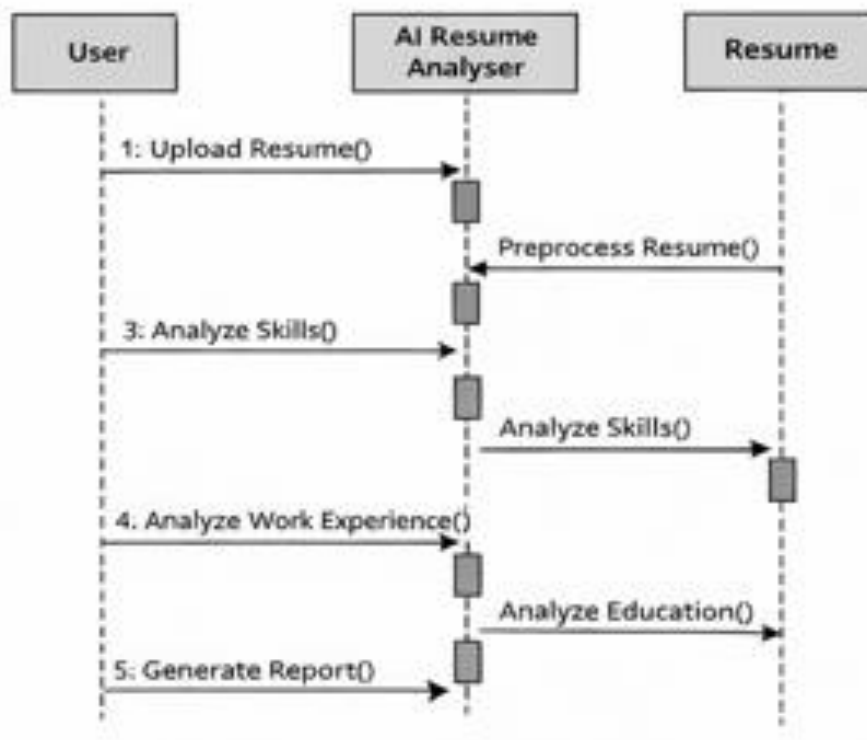
CLASS DIAGRAM



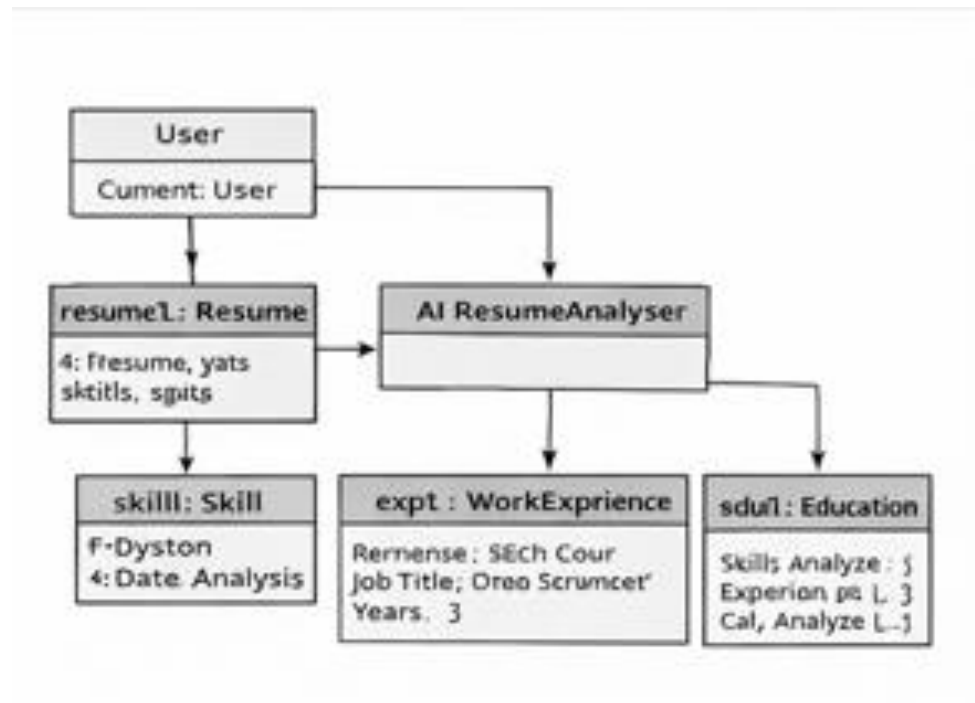
ACTIVIY DIAGRAM –



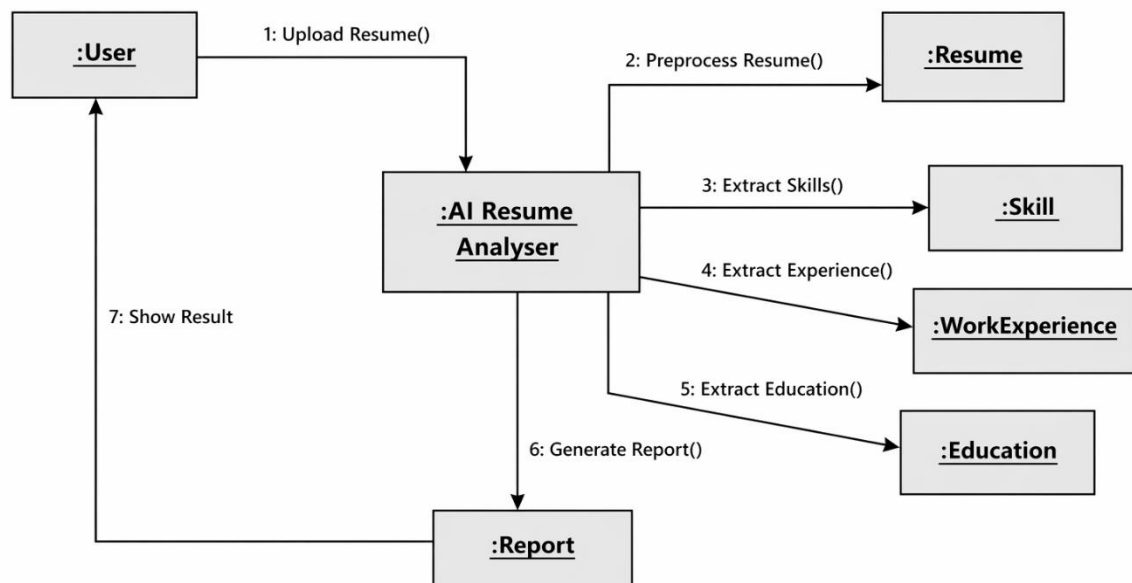
SEQUENCE DIAGRAM –



OBJECT DIAGRAM –

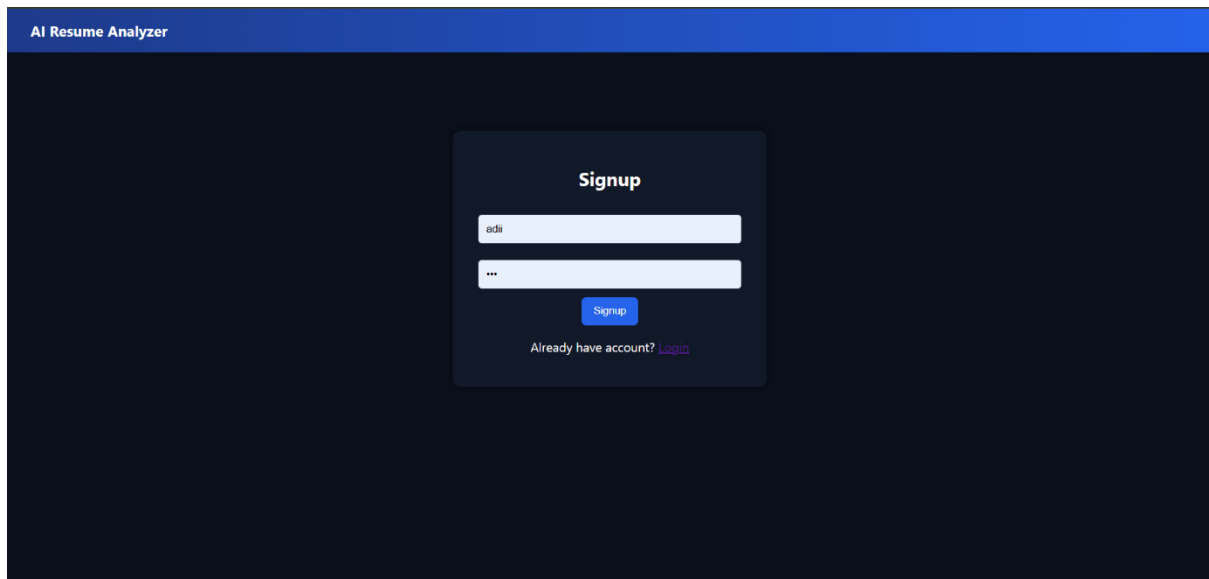


COLLABORATION DIAGRAM



❖ SYSTEM INTERFACE -

SIGNUP PAGE –



The screenshot shows the 'Signup' page of the 'AI Resume Analyzer'. The page has a dark blue header with the text 'AI Resume Analyzer'. The main content area is dark blue. In the center, there is a light blue rounded rectangle containing the 'Signup' form. The form has two input fields: the first contains 'adii' and the second contains three asterisks. Below the fields is a blue 'Signup' button. At the bottom of the form, it says 'Already have account? [Login](#)'.

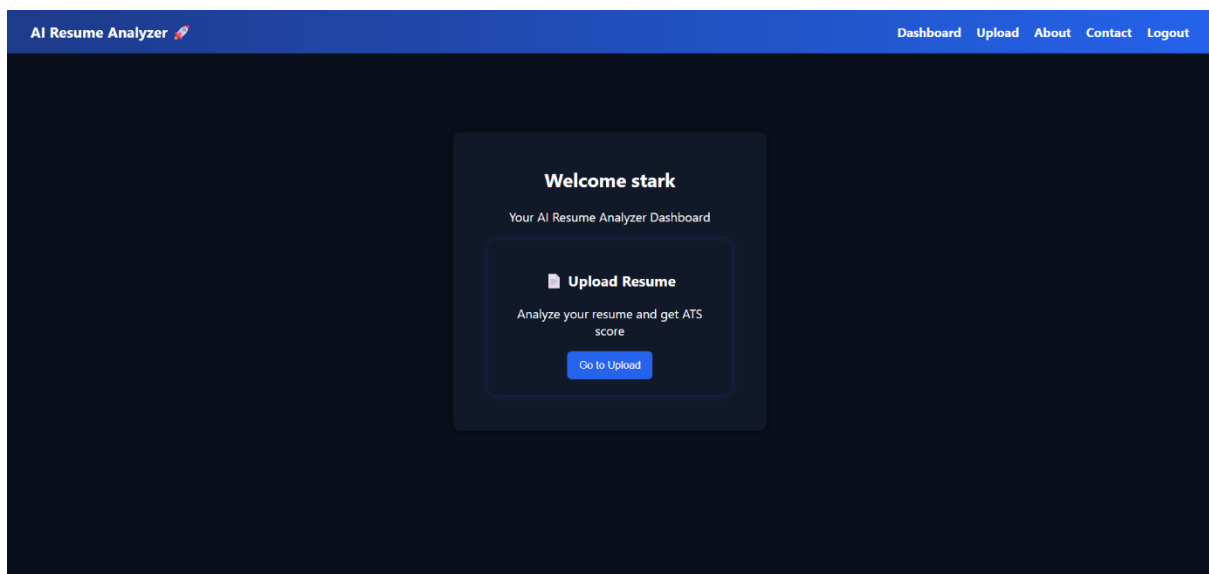
AI Resume Analyzer

Signup

[Signup](#)

Already have account? [Login](#)

DASHBOARD PAGE –



The screenshot shows the 'Dashboard' page of the 'AI Resume Analyzer'. The page has a dark blue header with the text 'AI Resume Analyzer' and a rocket icon. On the right side of the header, there are links: 'Dashboard', 'Upload', 'About', 'Contact', and 'Logout'. The main content area is dark blue. In the center, there is a light blue rounded rectangle containing the 'Welcome stark' message. Below the message, it says 'Your AI Resume Analyzer Dashboard'. There is a blue button labeled 'Upload Resume' with a document icon. Below the button, it says 'Analyze your resume and get ATS score'. At the bottom of the rectangle, there is a blue button labeled 'Go to Upload'.

AI Resume Analyzer 🚀

[Dashboard](#) [Upload](#) [About](#) [Contact](#) [Logout](#)

Welcome stark

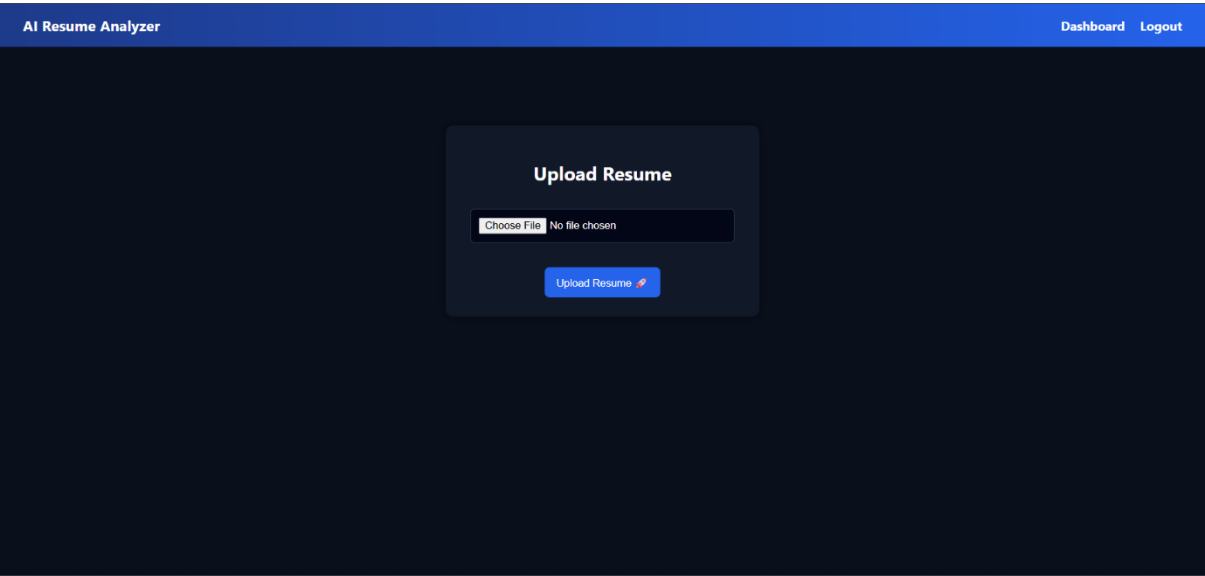
Your AI Resume Analyzer Dashboard

[Upload Resume](#)

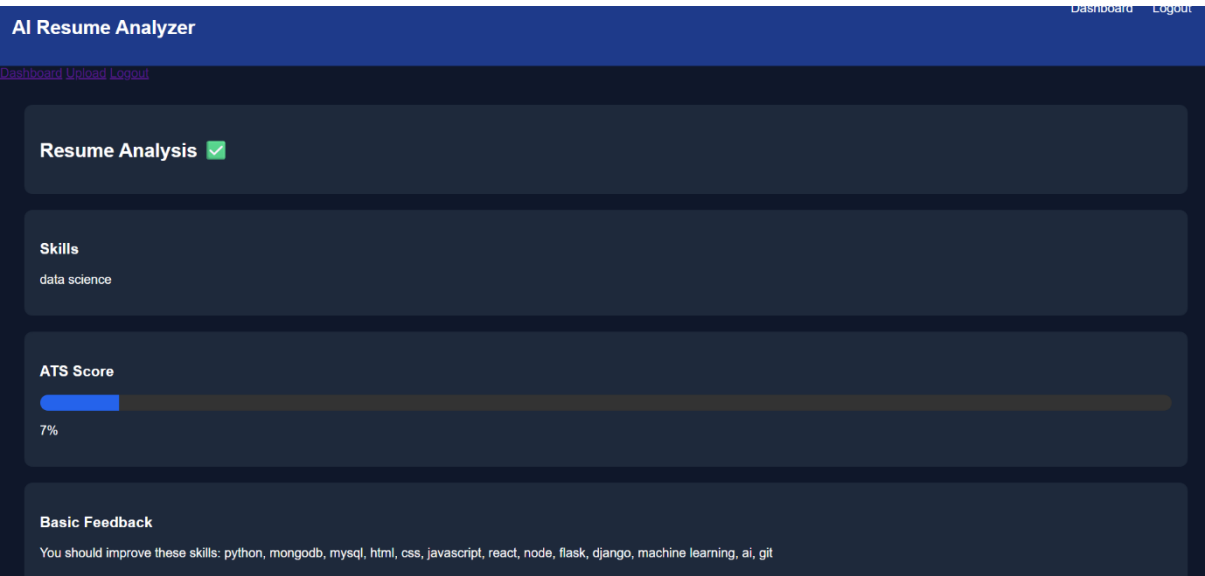
Analyze your resume and get ATS score

[Go to Upload](#)

RESUME UPLOAD SECTION –



RESUME ANALYSIS SECTION –



RESUME ANALYSIS SECTION –

AI Feedback

⚠️ Add more technical skills

❌ Low ATS score

★ Recommended skills: python, mongodb, mysql, html, css, javascript, react, node, flask, django, machine learning, ai, git

💡 Add projects and achievements

💡 Improve formatting and keywords

📌 Recommended Jobs:

Software Developer

Recommended Jobs 📌

Software Developer

Download Report 📄

Download PDF

ADMIN PAGE –

Admin Panel

UsersAnalyticsLogout

Total Users

6

Admins

1

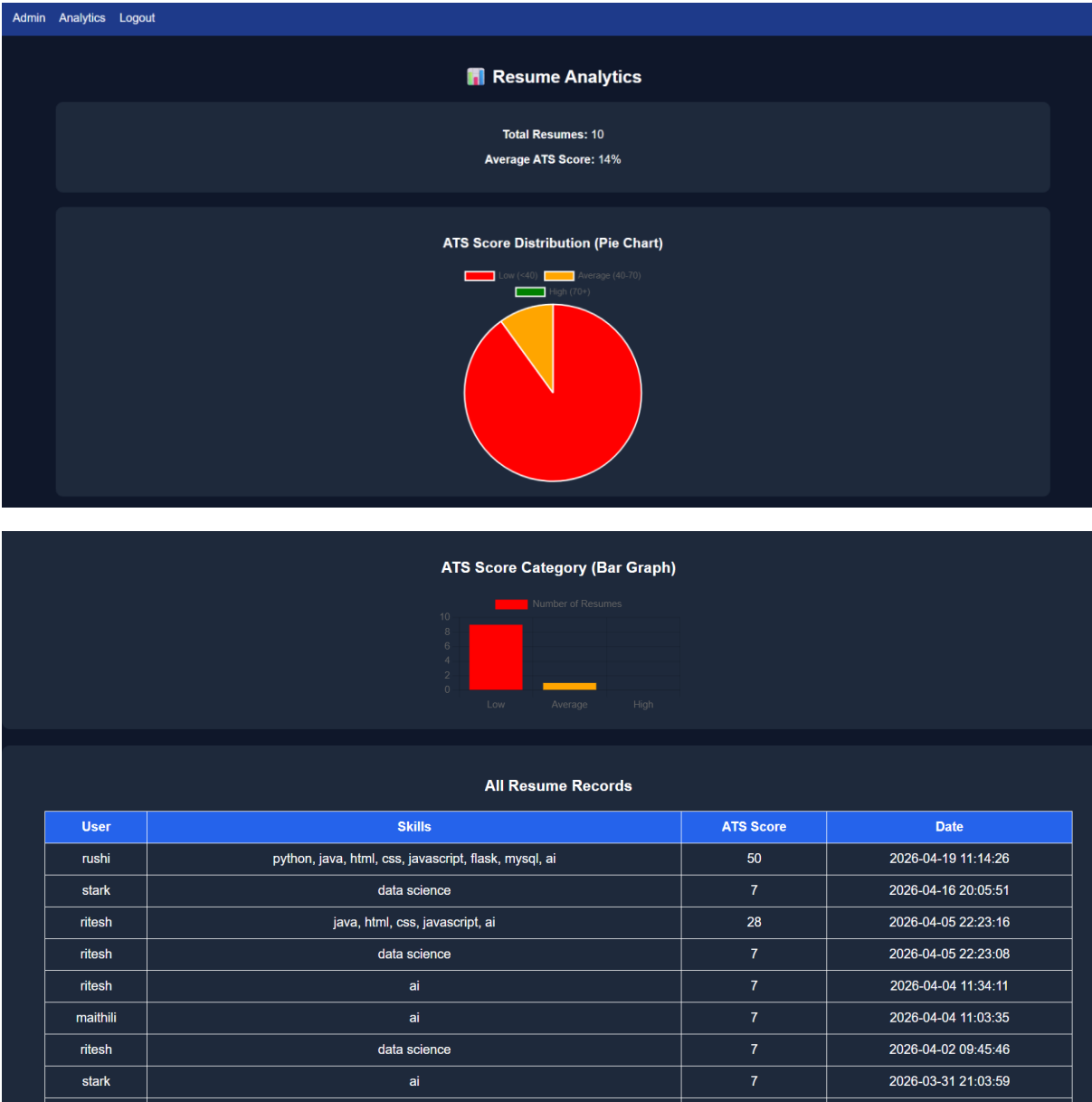
Users

5

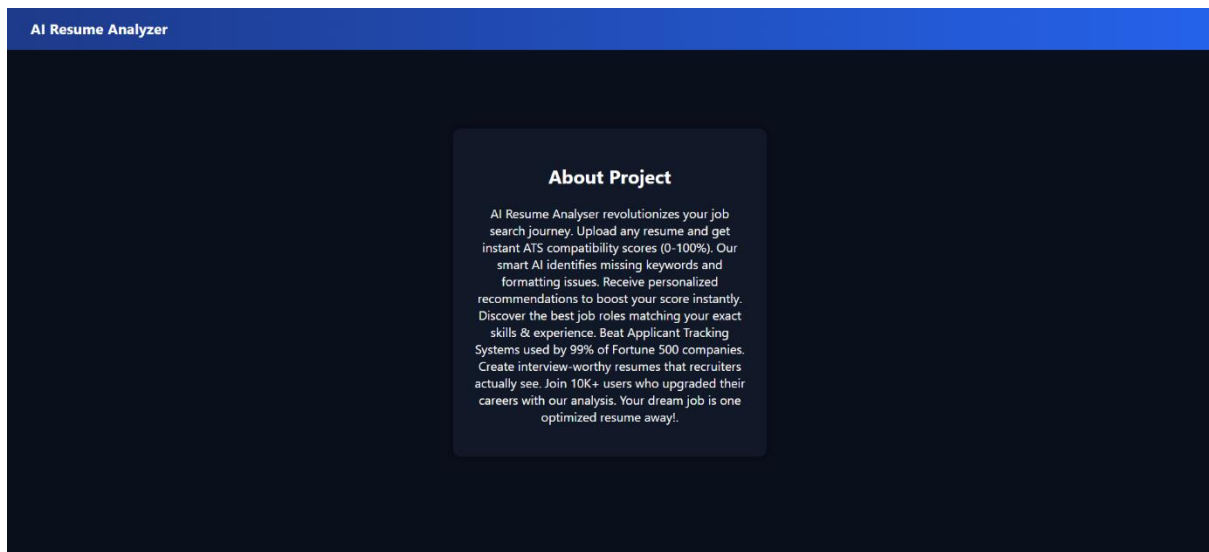
User Management

ID	Username	Role	Actions	
2	adli	Admin	Delete	Make User
6	alok	User	Delete	Make Admin
7	stark	User	Delete	Make Admin
8	rushi	User	Delete	Make Admin
9	ritesh	User	Delete	Make Admin
10	maithili	User	Delete	Make Admin

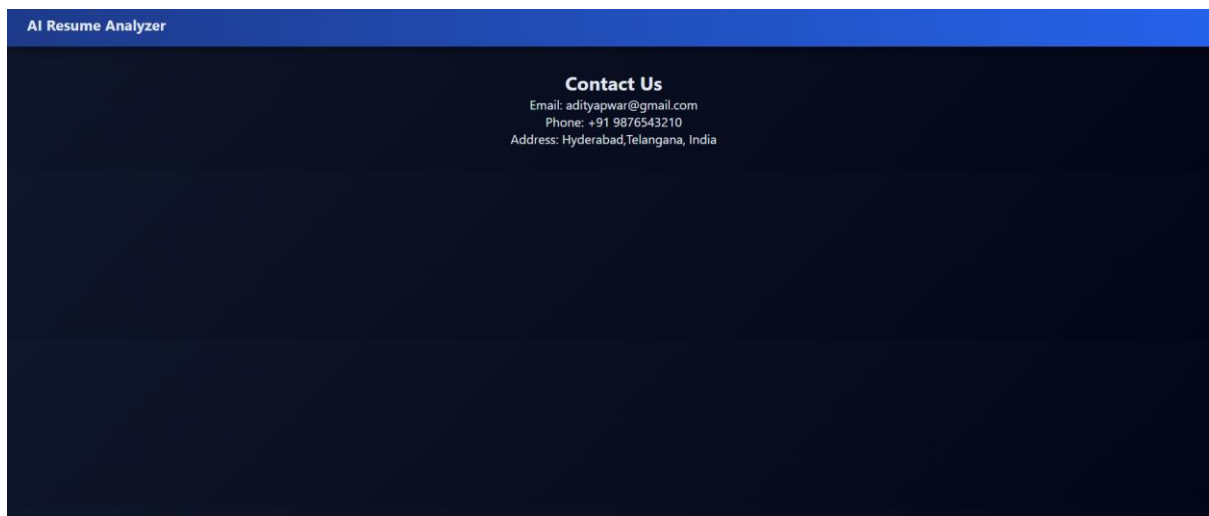
ATS SCORE CHECKER –



ABOUT SECTION –



CONTACT US PAGE –



❖ DATA DICTIONARY –

USER

Table Name: user				
Primary Key:		Id no		
Foreign Key:				
Table Description		It stores details of user which signup website		
Sr.no	Field Name	Data Type	Constraint	Description
1.	id	Int	Primary Key	Id of user
2.	username	Varchar (100)	Unique Key	Name of user
3.	password	Varchar (100)	Not Null	Password of user
4.	Role	Varchar(100)	Current time	Role(admin/user)

RESUMES

Table Name: Resumes				
Table Description		It stores details of resumes		
Sr.no	Field Name	Data Type	Constraint	Description
1.	id	Int	Primary Key	Id of user
2.	username	Varchar (100)	Unique Key	Name of user
3.	skills	TEXT	Not Null	Extract skills from resume
4.	ats_score	Int	-	ATS Score (%)
5.	created_at	TIMESTAMP	-	Update for date and time

❖ IMPLEMENTATION DETAILS

Client-Side:

- Processor: AMD Ryzen 7 5700U with Radeon Graphics
- RAM: 16GB.
- Hard Disk: 512GB.
- Devices: Computer, Laptop etc...
- Browsers: Latest upgraded any browser like Chrome, opera, Safari etc.

Server-Side:

- Processor: AMD Ryzen 7 5700U with Radeon Graphics 1.80 GHz
- RAM: 16
- Hard Disk: 512GB
- Data Base Server: PhpMyAdmin
- Language: PHP 3.3.0

Development-Side :

- Client Side: HTML5, CSS3, JavaScript
- Sever Side Language: Python 3.13 with Flask 2.x
- Data Base Server: XAMPP SERVER(My SQL)
- Operating System: Windows 10 or above
- Editors: Visual Code and Notepad
- Browser: Latest upgraded any browser like Chrome, Chrome, Safari, Internet Explorer, Opera etc.

❖ OUTPUT AND REPORT TESTING

The AI Resume Analyzer System is tested to ensure all functionalities work correctly and efficiently. A test plan is created to verify modules like login, resume upload, and analysis. Black box testing is used to check system outputs for different inputs without knowing internal logic, while data validation ensures only valid files are accepted. White box testing is performed to verify internal working such as data extraction and scoring. Various test cases are executed to check expected results, like correct scoring for valid resumes and error messages for invalid inputs.

- TEST PLAN :

The test plan for the AI Resume Analyzer System is designed to ensure that all modules of the system function correctly and meet user requirements. It includes testing of features such as user registration and login, resume upload, data extraction, resume analysis, scoring, and suggestion generation. Different types of inputs, including valid and invalid resumes, are used to verify system behaviour. The plan focuses on checking accuracy, performance, and reliability of the system. Each module is tested individually as well as in integration to ensure smooth workflow. The testing process helps in identifying errors and ensuring that the system provides correct and efficient results.

- Black Box Testing or Data Validation Test Cases

Black box testing is performed to verify the functionality of the **AI Resume Analyzer System** without considering the internal code structure. In this testing, various inputs such as valid and invalid resumes are provided to check whether the system produces correct outputs. Data validation ensures that only supported file formats like PDF and DOC are accepted, and incorrect or empty inputs are rejected with proper error messages. The system is also tested for correct extraction of data, accurate scoring, and suggestion generation. This testing helps ensure that the system behaves as expected from the user's perspective and handles all input cases effectively.

- White Box Testing or Functional Validation Test Cases and Results

White box testing is performed to verify the internal logic and working of the AI Resume Analyzer System. It checks whether all functions such as resume parsing, data extraction, keyword matching, and scoring are working correctly. Functional validation ensures that each module of the system performs as expected according to the requirements. Test cases include checking correct data extraction from resumes, proper score generation, and accurate suggestions. The results show that valid inputs produce correct outputs, while invalid inputs are handled with appropriate error messages. This testing confirms that the system is reliable, accurate, and functions efficiently.

TEST CASES -

Upload Resume (Valid)	Upload PDF/DOC resume	Resume uploaded successfully	Pass
Upload Resume (Invalid File)	Upload unsupported file (e.g., .jpg)	Shows error "Invalid format"	Pass
Upload Resume (Empty)	Click upload without file	Shows warning message	Pass
Resume Parsing	Upload resume	System extracts text correctly	Pass
Skill Extraction	Upload resume	Skills identified from resume	Pass
Keyword Matching	Upload resume + job role	Matching keywords highlighted	Pass
Resume Score Generation	Upload resume	Score displayed based on content	Pass
Missing Skills Detection	Analyze resume	Shows missing skills for job role	Pass
Suggestions/Improvements	Analyze resume	Shows tips to improve resume	Pass
Download Report	Click download	Report downloaded successfully	Pass
Multiple Uploads	Upload another resume	Previous data replaced with new	Pass
Performance Check	Upload large resume	System processes without crash	Pass
Error Handling	Upload corrupted file	Shows proper error message	Pass
User Interface	Use all features	Smooth and responsive UI	Pass

❖ CONCLUSION

The **AI Resume Analyzer System** is a web-based application that automates the process of resume screening and evaluation efficiently. It allows users to upload resumes and receive instant analysis, scoring, and improvement suggestions, while admins can manage job roles and required skills. The system reduces manual effort, saves time, and minimizes errors in the recruitment process.

Developed using Python (Flask), MySQL, HTML, CSS, JavaScript, and Bootstrap, the system is user-friendly and ensures proper data management and accurate analysis. The automated evaluation and feedback feature helps candidates improve their resumes according to industry standards.

Overall, the system provides a smart, fast, and reliable solution for resume analysis and is useful for both recruiters and job seekers, helping improve efficiency and decision-making in the hiring process.

❖ RECOMMENDATIONS:

The AI Resume Analyzer System can be further enhanced by integrating advanced Machine Learning and Natural Language Processing techniques to improve the accuracy of resume analysis. The system can be extended to support multiple languages, making it more accessible to a wider range of users.

Integration with online job portals can make the system more practical for real-world recruitment processes. Additional features like detailed analytics, personalized suggestions, and real-time recruiter interaction can improve user experience.

Improving the user interface and adding more customization options will make the system more attractive and easier to use. Regular updates and maintenance are recommended to keep the system efficient, secure, and up to date with industry requirements.

- **FUTURE SCOPE**

The **AI Resume Analyzer System** has been developed to overcome the limitations of the manual resume screening process and provide an automated, efficient solution. This web-based application reduces human effort, minimizes errors, and provides accurate analysis with a user-friendly interface. The system is designed in a modular way so that future enhancements and modifications can be easily implemented without affecting the existing functionality.

There is always scope for improvement in any software system. The current system can be enhanced by integrating advanced technologies such as Machine Learning and Deep Learning to improve the accuracy and understanding of resume content. The application can also be deployed on the internet and integrated with online job portals to make it more practical and widely accessible.

Further improvements can include adding more features to reduce admin workload and automate job role management. The system can be extended to support multiple languages and provide more detailed analytics and personalized suggestions. Security features can also be enhanced to protect sensitive user data.

We plan to implement the following enhancements in the future:

- Use of advanced AI techniques for better resume analysis
- Integration with job portals and recruitment platforms
- Addition of more automated features to reduce admin work
- Support for multiple languages
- Improved user interface and advanced analytics features

❖ BIBLIOGRAPHY:

- Web Programming using HTML, CSS, and JavaScript
- Python Programming for Web Development (Flask)
- Database System Concepts
- Software Engineering – Project Development and Documentation
- Introduction to Artificial Intelligence and NLP
- Flask Official Documentation
- Bootstrap Official Documentation
- W3Schools Tutorials for HTML, CSS, JavaScript, and MySQL
- Online Resume Analyzer project references and documentation

❖ REFERENCES

- “Core JavaScript ”- by R. Nageswara Rao
- “A Byte of Python”-by C.H.Swaroop
- “Taming JavaScript”-by Jeeva Jose
- “CSS crash course”-by Eric Matthes
- “CSS Pocket Reference”-by Mark Lutz
- “Database Management System”-by Ramakrishnan Gherkin • • “Database Management System”- by Ashwini S Diwakar.
- “Software Engineering”- by Ashwini S Diwakar.